EXERCISE 1 ; SQL FUNDAMENTALS

1. SELECT STATEMENT

```
SELECT *
FROM employees;
```

id	first_name	last_name	department	salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

2. SELECT DISTINCT STATEMENT

```
SELECT DISTINCT department
FROM employees;
```

department
IT
HR
finance
Marketing

3. ORDER BY STATEMENT

```
SELECT first-name, last-name
FROM employees
ORDER BY salary DESC;
```

first-name	last-name	salary
Emily	~~Doe~~ Davis	62000
mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Daniel	Clark	53000
Sarah	Brown	53000
David	White	52000
Jessica	moore	51000
Laura	Hall	50000
Jane	Smith	48000

4. Limit STATEMENT

```
SELECT *
FROM employees
ORDER BY salary DESC
LIMIT 5;
```

id	first-name	last-name	department	salary	hire-date	city
6	Emily	Davis	IT	62000	2015-02-14	Chicago
3	mike	Johnson	finance	60000	2017-09-30	Los Angeles
7	Robert	Wilson	finance	59000	2019-10-01	Houston
1	John	Doe	IT	55000	2018-06-15	New York
9	Daniel	Clark	marketing	53000	2022-06-01	Chicago

5. WHERE STATEMENT

```
SELECT *
FROM employees
WHERE department = 'IT';
```

id	first-name	last-name	department	salary	hire-date	city
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

6. AND STATEMENT

```
SELECT *
FROM employees
WHERE department = 'finance'
AND salary > 58000;
```

id	first-name	last-name	department	salary	hire-date	city
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
7	Robert	Wilson	finance	59000	2019-10-01	Houston

7. OR STATEMENT

```
SELECT *
FROM employees
WHERE department = ~~'HR'~~ 'HR'
OR department = 'marketing';
```

id	first-name	last-name	department	salary	hire-date	city
2	Jane	Smith	HR	48000	2019-07-20	Chicago
5	David	White	marketing	52000	2016-04-10	San Francisco
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	marketing	53000	2022-08-10	Chicago

8. NOT STATEMENT

```
SELECT *
FROM employees
WHERE department NOT = 'IT';
```

id	first-name	last-name	department	Salary	hire-date	City
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	mike	Johnson	Finance	60000	2017-09-30	Los Angeles
5	David	White	marketing	52000	2016-04-10	San Francisco
7	Robert	Wilson	finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

9. IN STATEMENT

```
SELECT * first-name, last-name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

First-name	last-name	department
John	Doe	IT
Jane	Smith	HR
mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	finance
Jessica	moore	HR
Laura	Hall	IT

10. COMBINING CONDITIONS

```
SELECT *
FROM employees
WHERE department = 'IT'
AND salary > 50000
AND city = 'NEW YORK'
```

id	first-name	last-name	department	salary	hire-date	city
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York

11. COMBINING WHERE, AND, and ORER BY

```
SELECT first-name, last-name
FROM employees
WHERE (department = 'finance' OR department = 'marketing')
AND salary > 52000
ORDER BY salary DESC;
```

first-name	last-name	department	salary
mike	Johnson	finance	60000
Robert	Wilson	finance	59000
Daniel	Clark	marketing	53000

12. COMBINING SELECT DISTINCT, WHERE and IN

```
SELECT DISTINCT city
FROM employees
WHERE department NOT IN ('IT', 'HR');
```

city
Los Angeles
San Francisco
Houston
Chicago

13. COMBINING WHERE, NOT, AND and ORDER BY

```
SELECT *
FROM employees
WHERE department NOT = 'finance'
AND salary > 50000
ORDER BY hire-date ASC;
```

id	first-name	last-name	department	salary	hire-date	city
6	Emily	Davis	IT	62000	2015-02-14	Chicago
5	David	White	marketing	52000	2016-04-10	San francisco
1	John	Doe	IT	55000	2018-06-15	New York
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
9	Daniel	Clark	marketing	53000	2022-06-01	Chicago

14. COMBINING WHERE, OR, IN and LIMIT

```
SELECT *
FROM employees
WHERE city IN ('Chicago' OR 'Los Angeles')
AND department IN ('IT' OR 'marketing')
LIMIT 3;
```

id	first-name	last-name	department	salary	hire-date	city
6	Emily	Davis	IT	62000	2015-02-14	Chicago
9	Daniel	Clark	marketing	53000	2022-06-01	Chicago